Sunton ESP32-8048S043C connectors (1.25 mm, 1:1 cables).
Check the cable maps pin 1 to pin 1 before power-up.

3.3 V for the scanner: the SparkFun header takes 3.3 V, not 5 V.

SparkFun 2D Barcode Scanner Breakout (DE2120, red aiming line).
Plugs in on top; M3 x 11 mm standoff in H1.
GM65 instead: J4. Fit only one scanner.

PWR_FLAG
+5V
PWR_FLAG
GND

J1
GND 1
SCAN_TX 2
SCAN_RX 3
+5V 4
Sunton P1 UART

AMS1117-3.3
+5V 3 VI
VO 2 +3V3
GND 1

C4 22uF +5V GND
C1 10uF +5V GND
C2 22uF +3V3 GND
C3 100nF +3V3 GND

J3
SCAN_STAT 1
SCAN_TRIG 2
+3V3 3
SCAN_TX 4
SCAN_RX 5
GND 6
SparkFun 2D Barcode Scanner

TP1 TRIG SCAN_TRIG
TP2 STAT SCAN_STAT

Status LED: WS2812B, data shifted 3.3 V -> 5 V (IO17).

J2
BUZZ 1
LED_3V3 2
3
GND 4
Sunton P4 IO

U2 74AHCT1G125
LED_3V3 2
+5V 5 VCC
GND 1 GND
4 LED_5V
GND 3
C5 100nF +5V GND

LED_5V
33R3
LED_DIN

D2 WS2812B
LED_DIN 4 DIN
+5V 1 VDD
DOUT 2 LED_DOUT
VSS 3 GND
C6 100nF +5V GND

TP3 LED_DOUT
LED_DOUT

Radar: HLK-LD2410C on a Dupont cable. OUT -> IO12,
UART on IO13 (RX) / IO11 (TX), 256000 baud.
IO11-13 are the Sunton SD pins: leave the SD slot empty.

Buzzer: MLT-8530 from 5 V, PWM on IO18 (ESPHome rtttl).

J6
IO13_RXD 1
IO12_PRESENCE 2
IO11_TXD 3
4
Sunton P2 IO

+5V 1
LS1 MLT-8530
BUZZ_SW 2

D1 1N4148W
+5V BUZZ_SW

RADAR_OUT
1kR5
IO12_PRESENCE

RADAR_TX
1kR6
IO13_RXD

IO11_TXD
1kR7
RADAR_RX

J5
+5V 1
GND 2
RADAR_OUT 3
RADAR_TX 4
RADAR_RX 5
LD2410C radar

BUZZ
100R1
BUZZ_G

BUZZ_G 1
Q1 IRLML6400A
2 GND
3 BUZZ_SW

BUZZ_G
100R2
GND

J4
GND 1
+5V 2
SCAN_TX 3
SCAN_RX 4
GM65 scanner

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H1 M3 standoff (scanner)
H3 M3 (case)
H4 M3 (case)

Sheet: /
File: scanner-carrier.kicad_sch

Title: Pantry scanner carrier

Size: A4	Date: 2026-10-27	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 1/1